

ENGINEERING MATERIAL SCIENCE: MASTER TEXTBOOK

1. Atomic Structure and Bonding

Primary Bonding Types

Materials properties are dictated by atomic bonds:

- **Ionic Bonding:** Electron transfer between metal and non-metal.
- **Covalent Bonding:** Sharing of electrons (e.g., Diamond, Polymers).
- **Metallic Bonding:** "Sea of electrons" (Metals).

2. Crystal Structures

Unit Cells

Most engineering metals crystallize in:

- **BCC (Body-Centered Cubic):** Higher strength, lower ductility (e.g., Iron at RT).
- **FCC (Face-Centered Cubic):** High ductility (e.g., Aluminum, Copper).

3. Mechanical Properties

Stress-Strain Behavior

Defined by Hooke's Law in the elastic region:

$$\sigma = E * \epsilon$$

where σ is stress, E is Young's Modulus, and ϵ is strain.

4. Phase Diagrams

Iron-Carbon System

Fundamental for understanding steel production. It shows the transformation of austenite to ferrite, cementite, and pearlite.

Problem 1: Tensile Testing

A steel bar with a 10mm diameter is subjected to a load of 50kN. If the extension is 0.5mm over a 200mm gauge length, calculate Stress and Strain.

Solution:

- Area (A) = $\pi * (5e-3)^2 \approx 7.85e-5 \text{ m}^2$.
- Stress (σ) = $F/A = 50e3 / 7.85e-5 \approx 637 \text{ MPa}$.
- Strain (ϵ) = $\Delta L / L = 0.5 / 200 = 0.0025$ (or 0.25%).